ECOCIDE

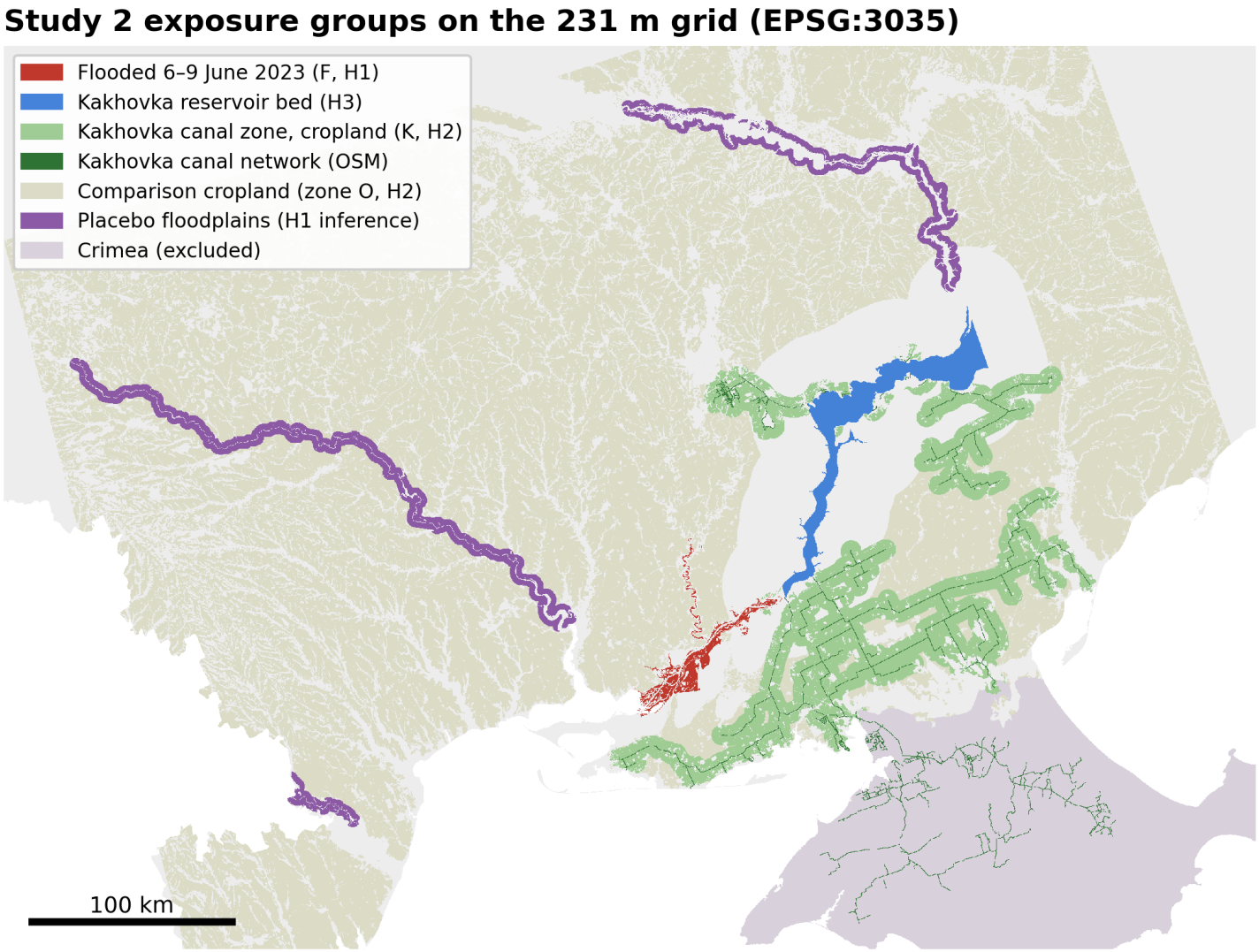
What did the destruction of the Kakhovka Dam do to vegetation?

Maps and Plots — Study 2 (main) and Study 1

Figure S2-1 — Study 2 exposure groups
Figure S2-2 — H1 flood: event studies
Figure S2-3 — H2 irrigation: event study (pre-registered)
Figure S2-4 — Randomization inference
Figure S2-5 — H3 former reservoir bed
Figure S2-6 — H4 decomposition of Kherson's change
Figure S2-7 — Registered revision: event studies
Figure S2-8 — Second revision: flood
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Figure S1-7 — Quarterly event study
Figure S1-8 — Primary and narrowed-baseline estimates with placebos
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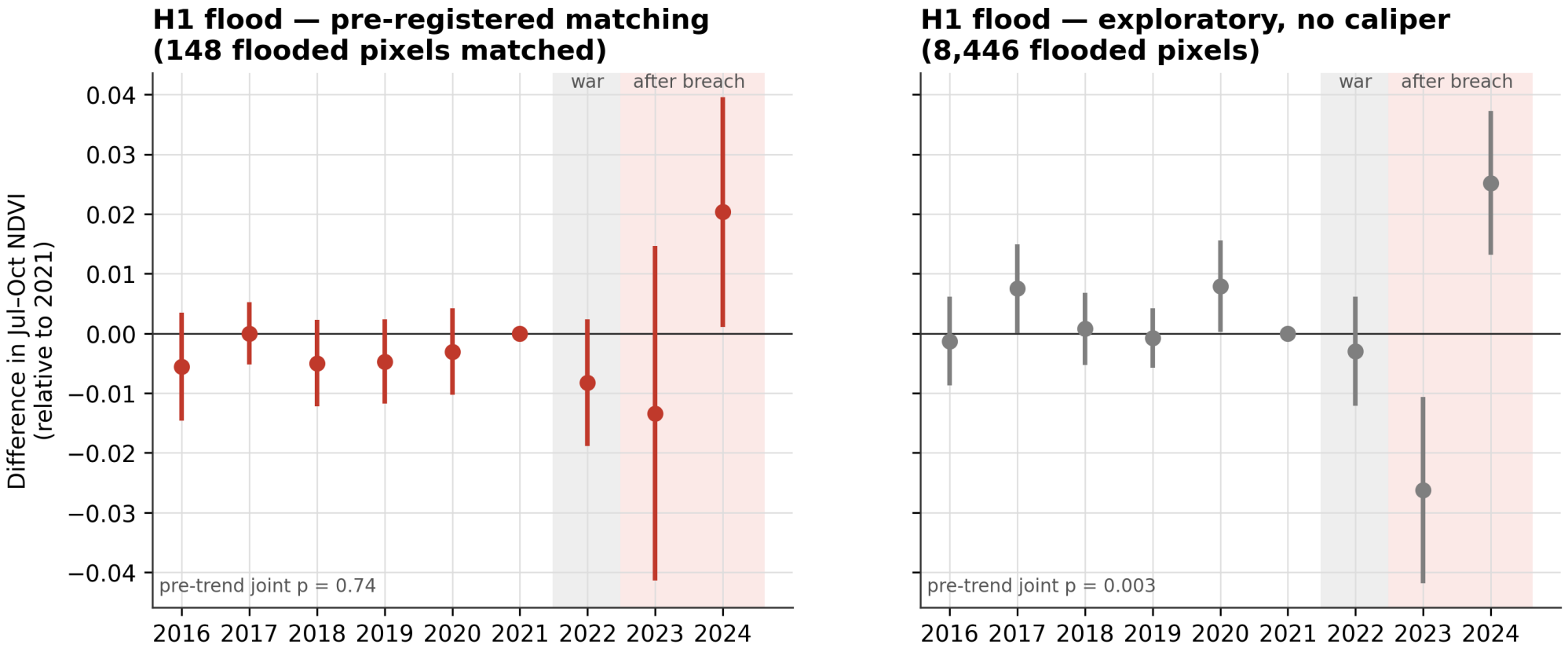
Sakshi D. Maske — Independent Geospatial Researcher

Figure S2-1 — Study 2 exposure groups



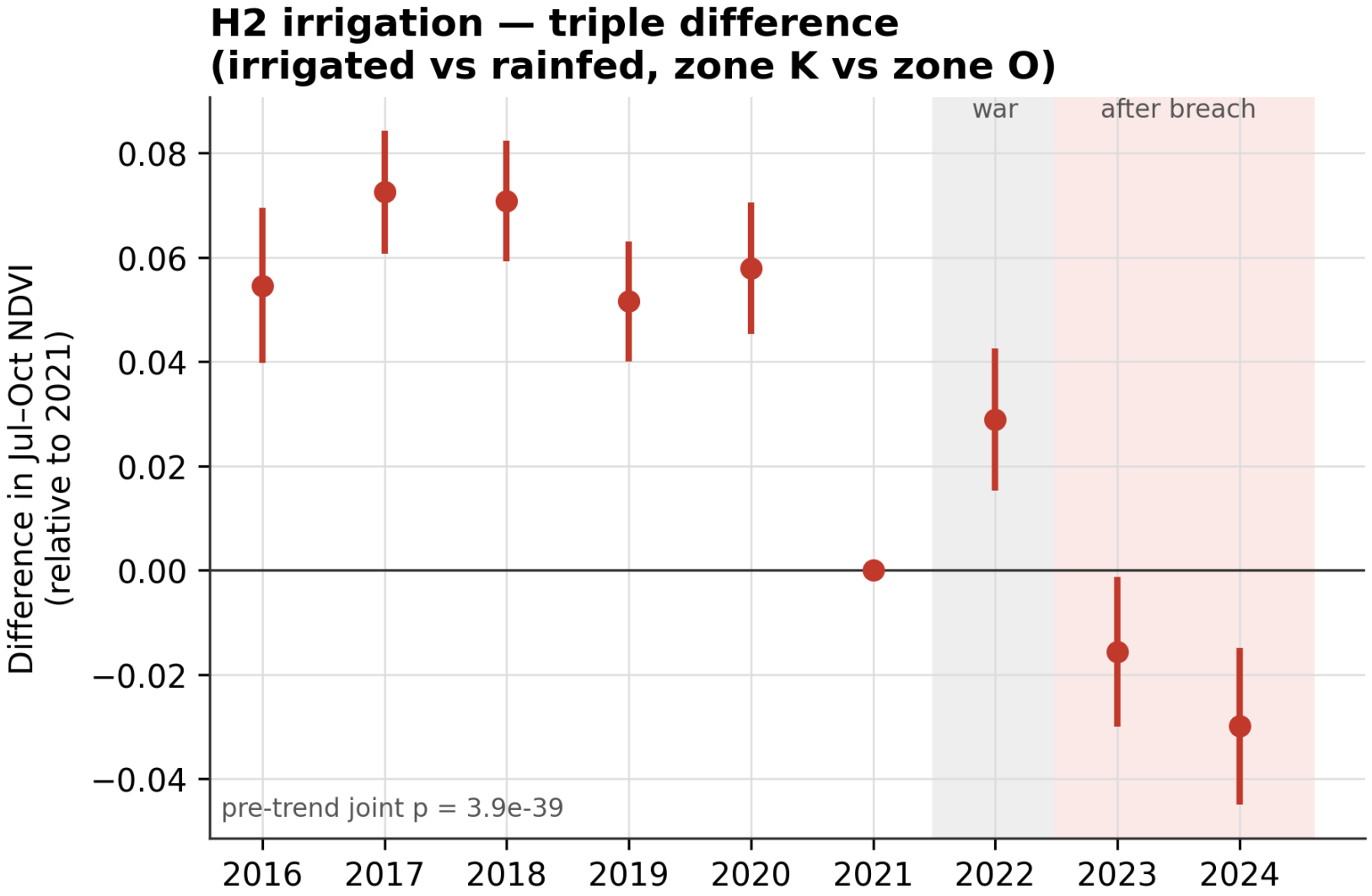
Flooded land (UNOSAT 6-9 June 2023 composite), former reservoir bed, Kakhovka canal zone (K) and comparison cropland (O), placebo floodplains; 231 m grid, EPSG:3035.

Figure S2-2 — H1 flood: event studies



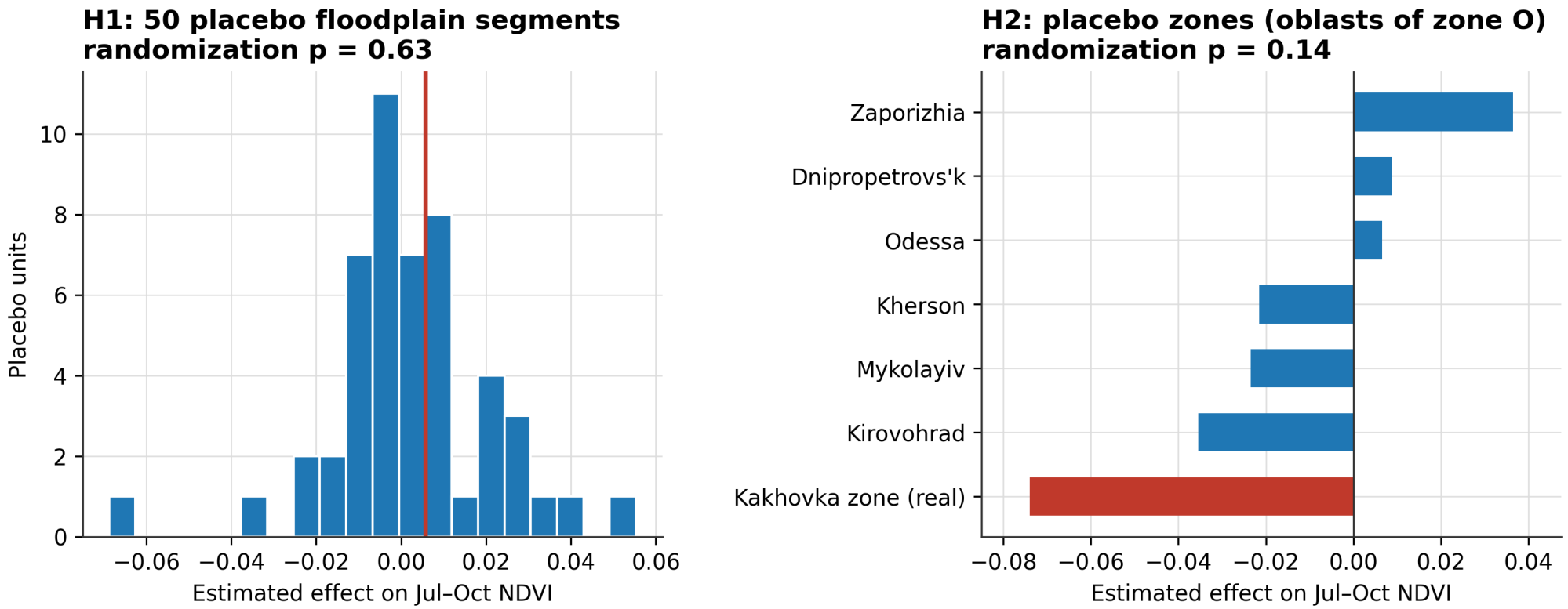
Flooded minus matched unflooded pixels, relative to 2021. Left: pre-registered caliper matching; right: exploratory matching without caliper. 95% cluster-robust CIs.

Figure S2-3 — H2 irrigation: event study (pre-registered)



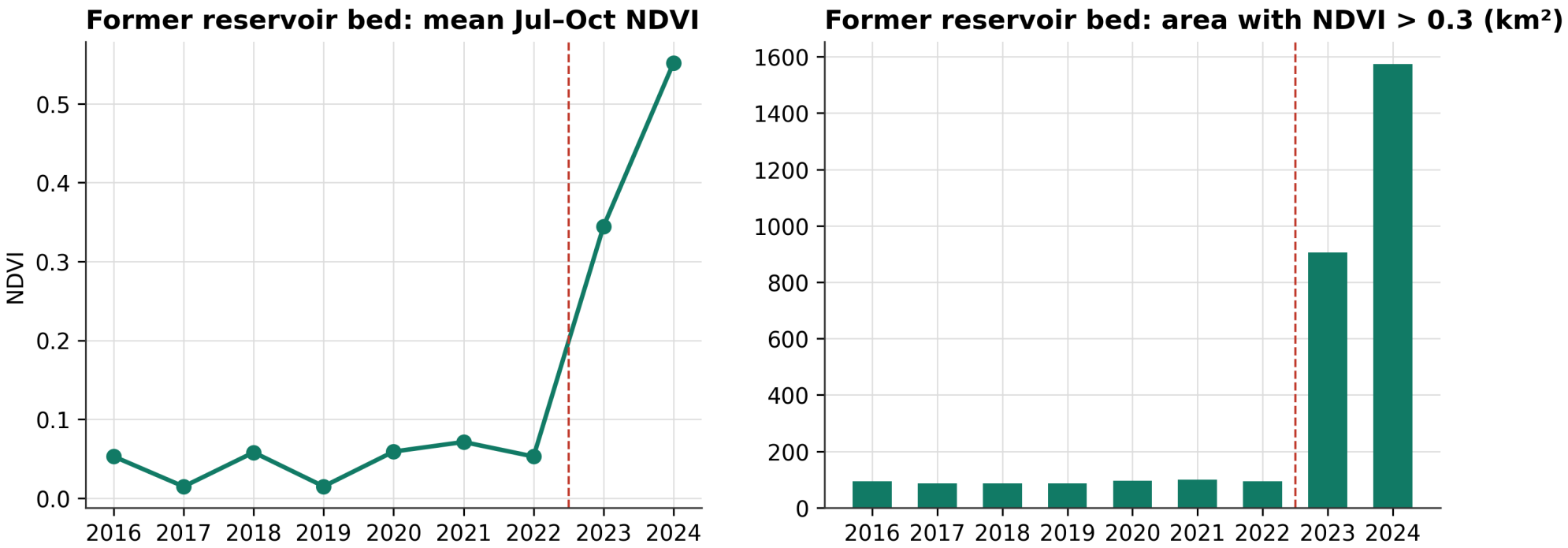
Irrigated minus rainfed cropland, canal zone minus comparison zone, relative to 2021. The reference year 2021 was the wettest July-October of 2016-2024 in the canal zone, and the contrast moves with summer rainfall, so post-breach values relative to 2021 cannot be read as 'no change'.

Figure S2-4 — Randomization inference



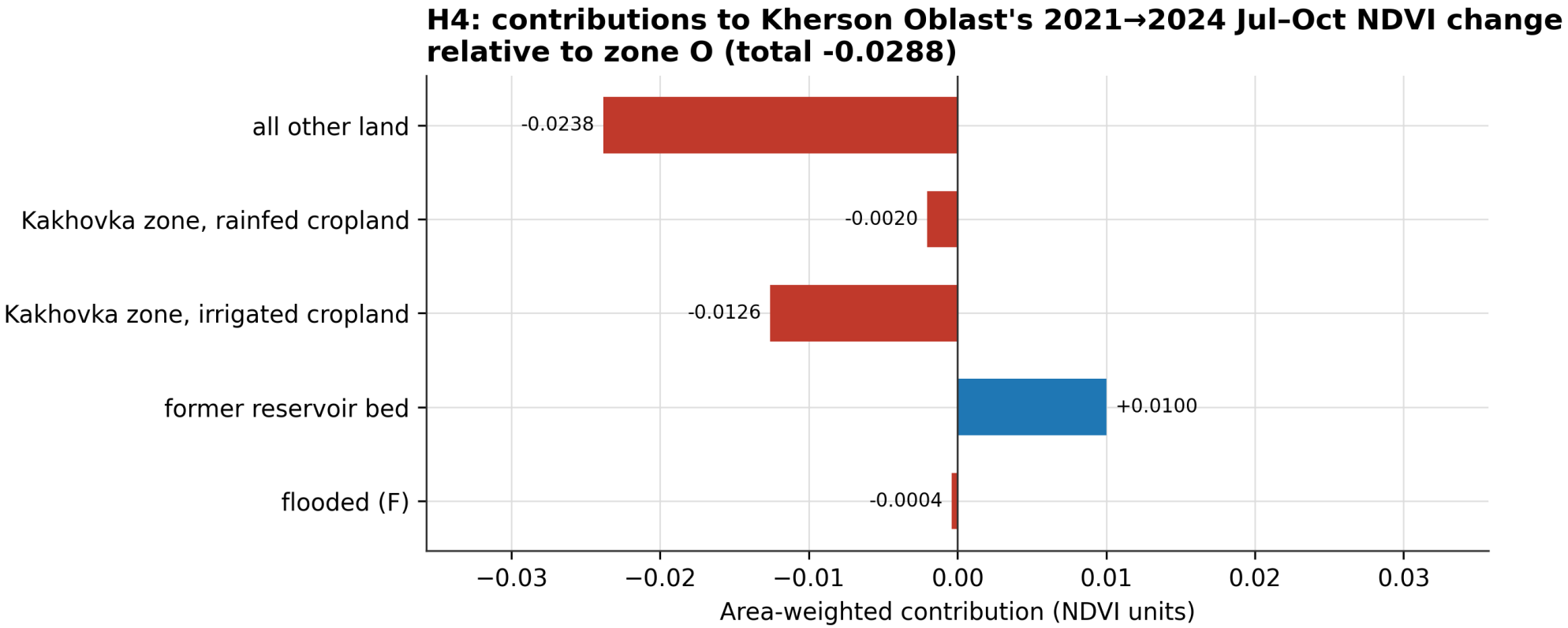
H1 estimate against 50 placebo floodplain segments; H2 estimate against six placebo oblasts.

Figure S2-5 — H3 former reservoir bed



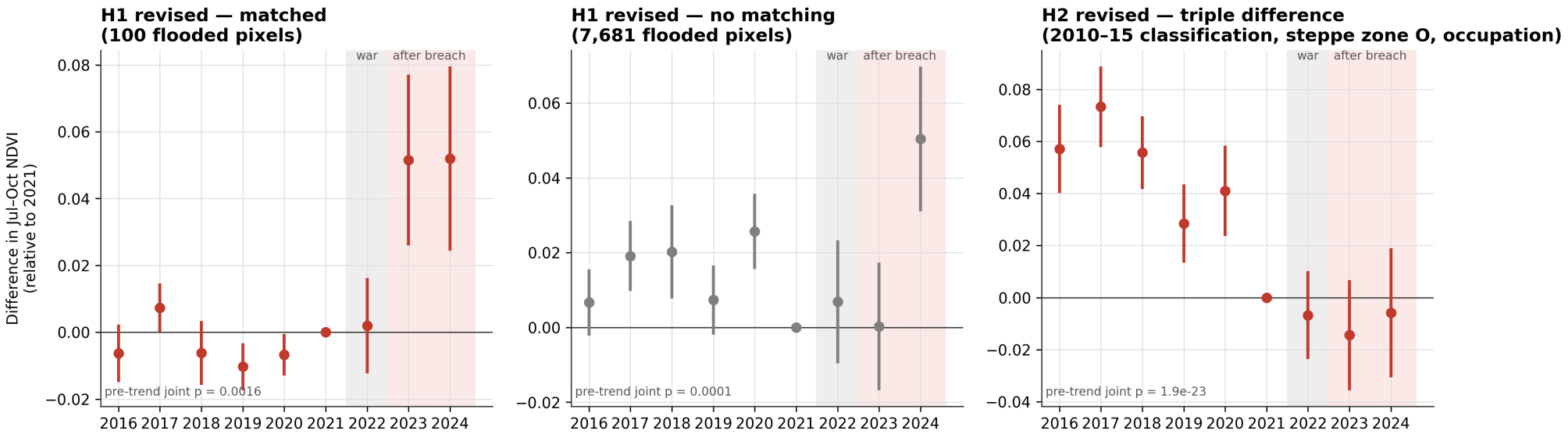
Mean July-October NDVI and area with NDVI above 0.3 on the former Kakhovka reservoir bed, 2016-2024.

Figure S2-6 — H4 decomposition of Kherson's change



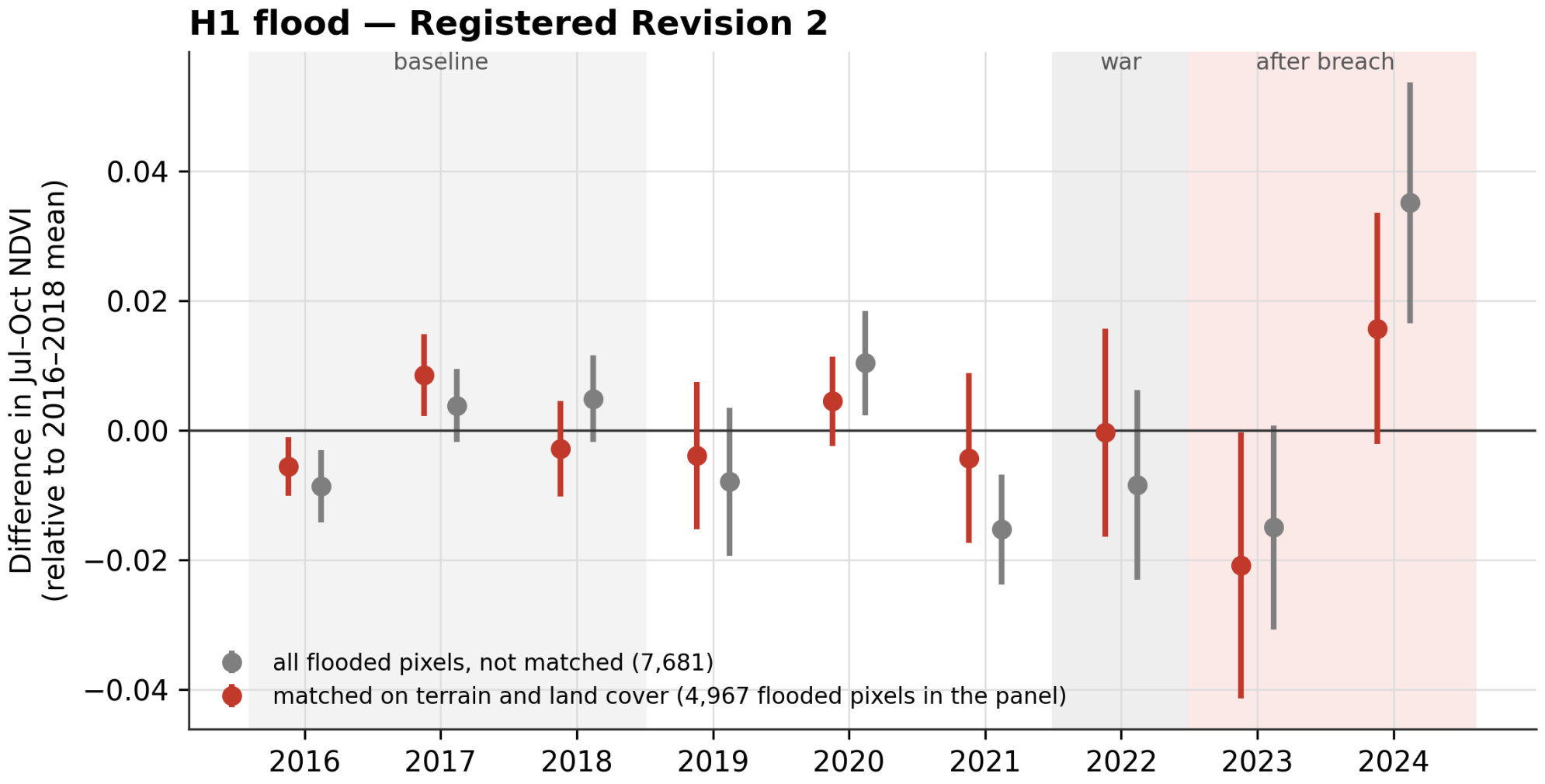
Area-weighted contributions to Kherson Oblast's 2021-2024 change in July-October NDVI relative to zone O. 'All other land' is land outside the exposure definitions; it includes irrigated land more than 5 km from the mapped canals. Descriptive, no uncertainty estimate.

Figure S2-7 — Registered revision: event studies



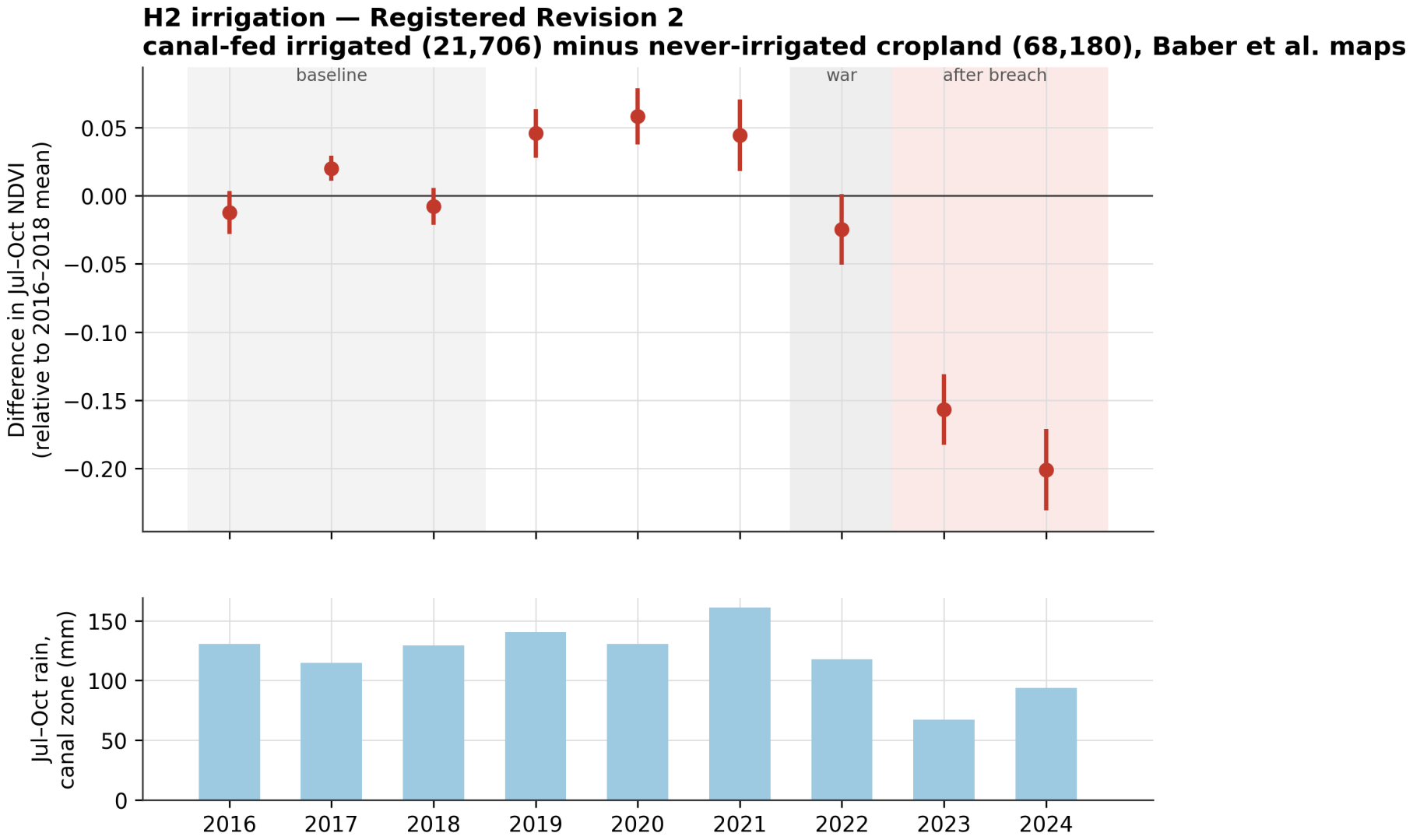
H1 matched and unmatched, and H2 triple difference under Registered Revision 1 (2010-15 irrigation classification, steppe comparison zone, VIINA occupation and conflict, annual water masks).

Figure S2-8 — Second revision: flood



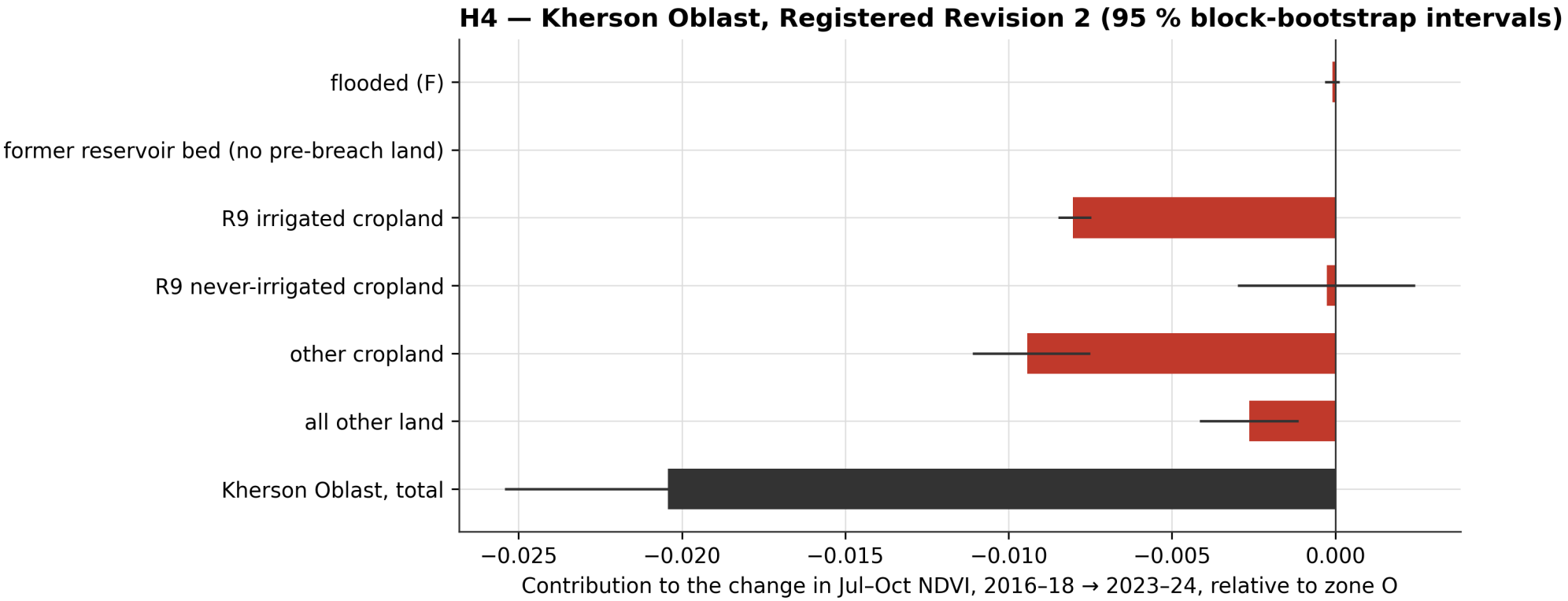
Flooded minus comparison land relative to 2016-2018: matched on height above the river and land cover (red) and not matched (grey). 95% CIs.

Figure S2-9 — Second revision: irrigation



Canal-fed irrigated minus never-irrigated cropland (Baber et al. 2026 maps) relative to 2016-2018, with July-October rainfall in the canal zone. 95% CIs.

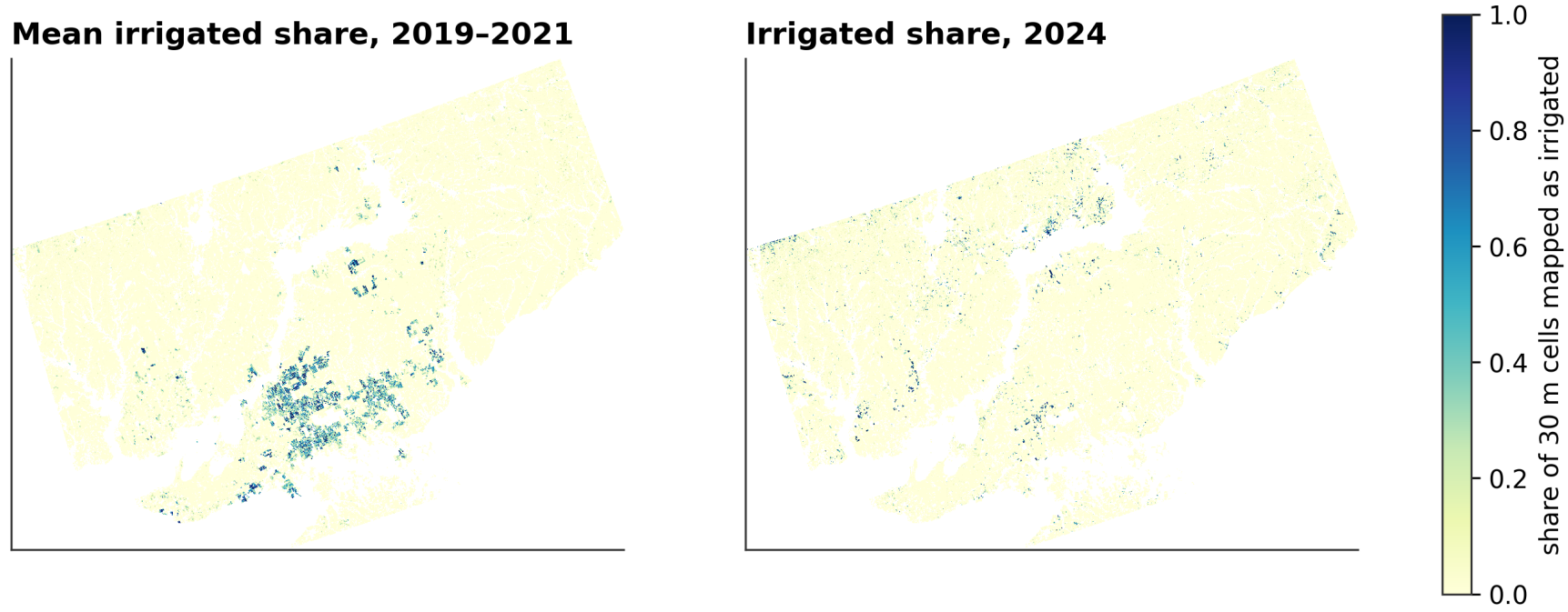
Figure S2-10 — Second revision: Kherson decomposition



Contributions to Kherson Oblast's change in July-October NDVI, 2016-18 to 2023-24, relative to the comparison area, with 95% block-bootstrap intervals.

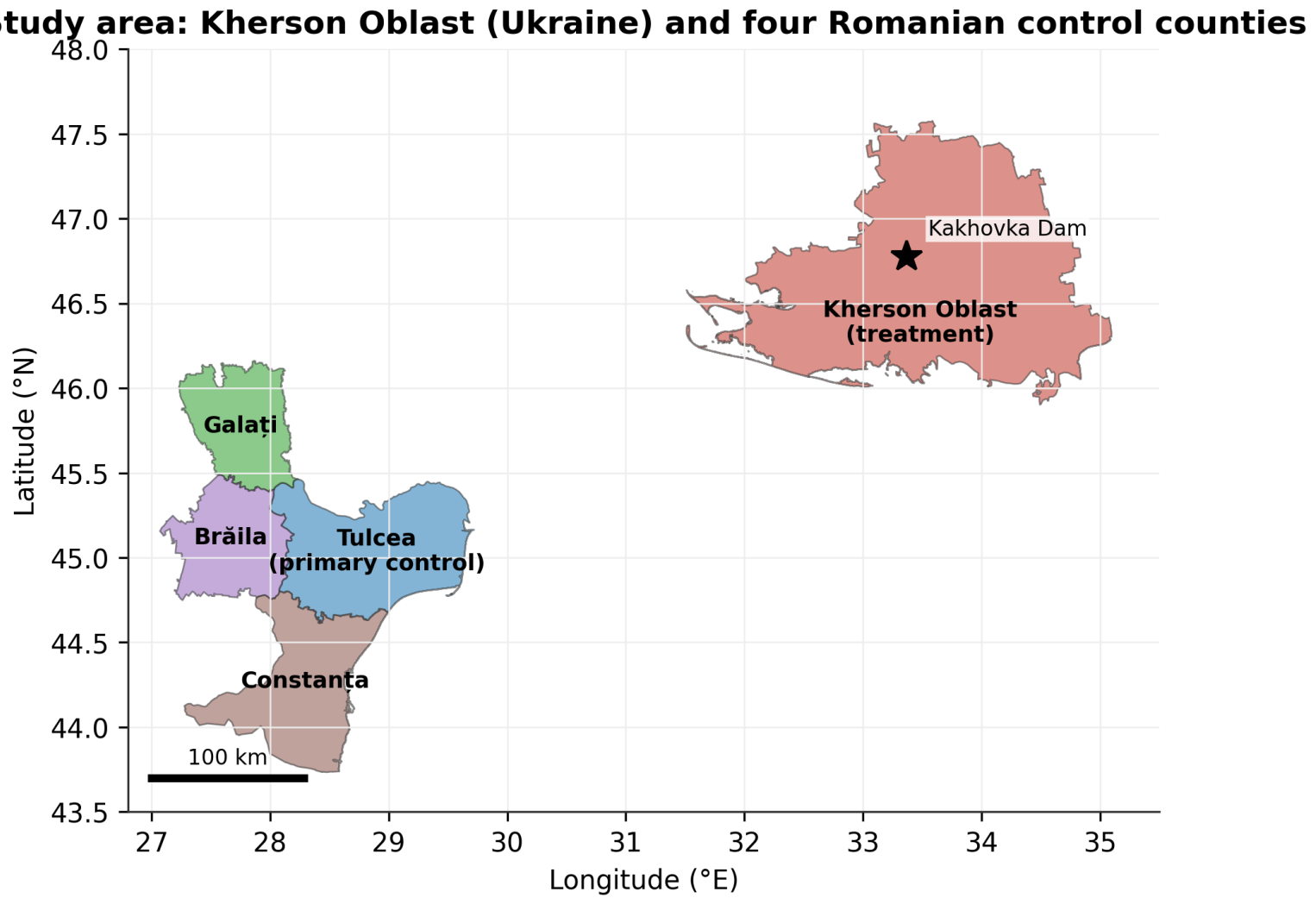
Figure S2-11 — Irrigated fields (Baber et al. 2026)

Irrigated fields of Baber et al. (2026b) on the 231 m grid, Kherson and Zaporizhzhia oblasts



Share of each 231 m pixel mapped as irrigated: mean 2019-2021 and 2024.

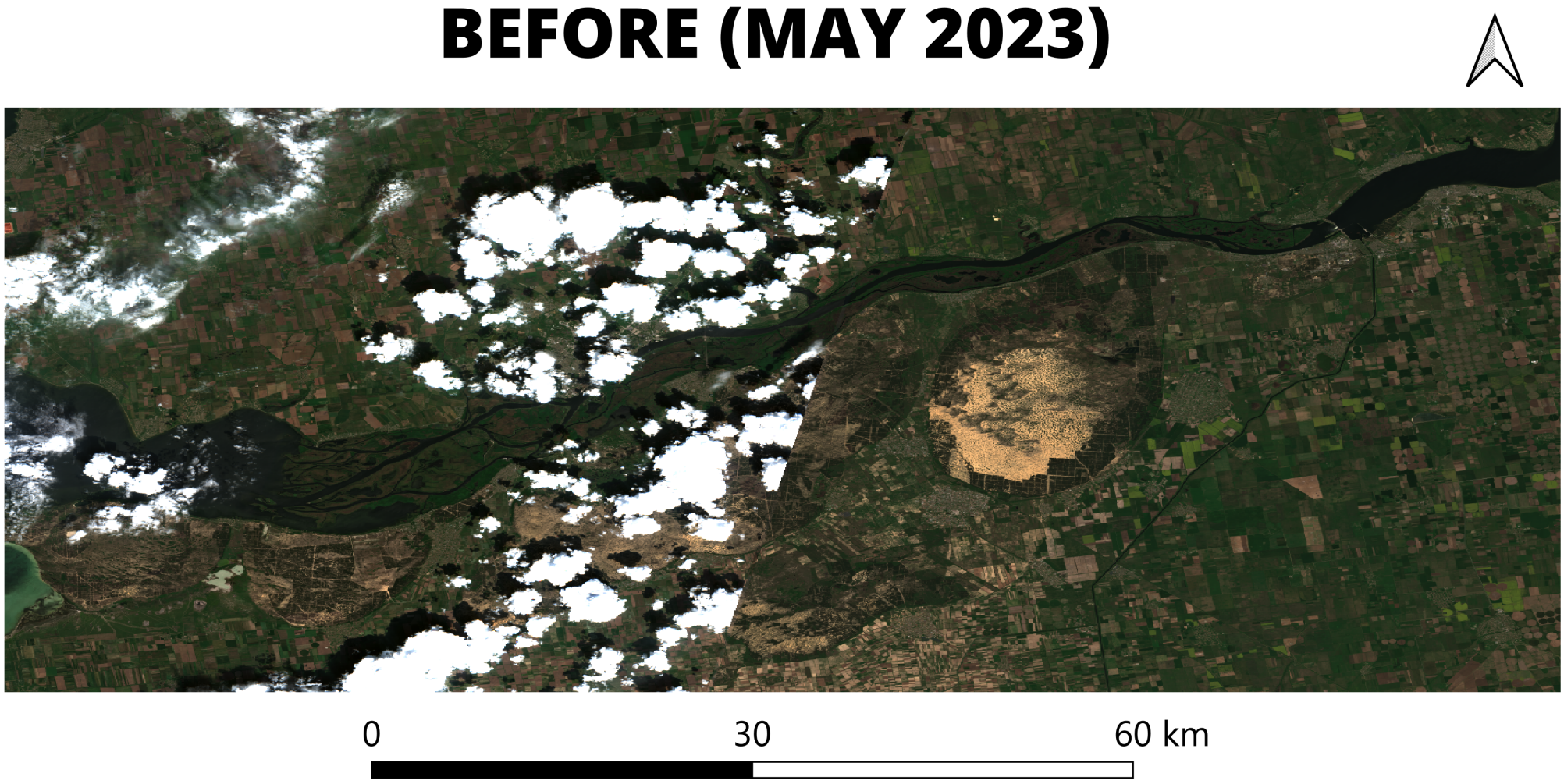
Figure S1-1 — Study area



Boundaries: GADM v4.1. Tulcea borders Ukraine's Odesa Oblast along the Danube.

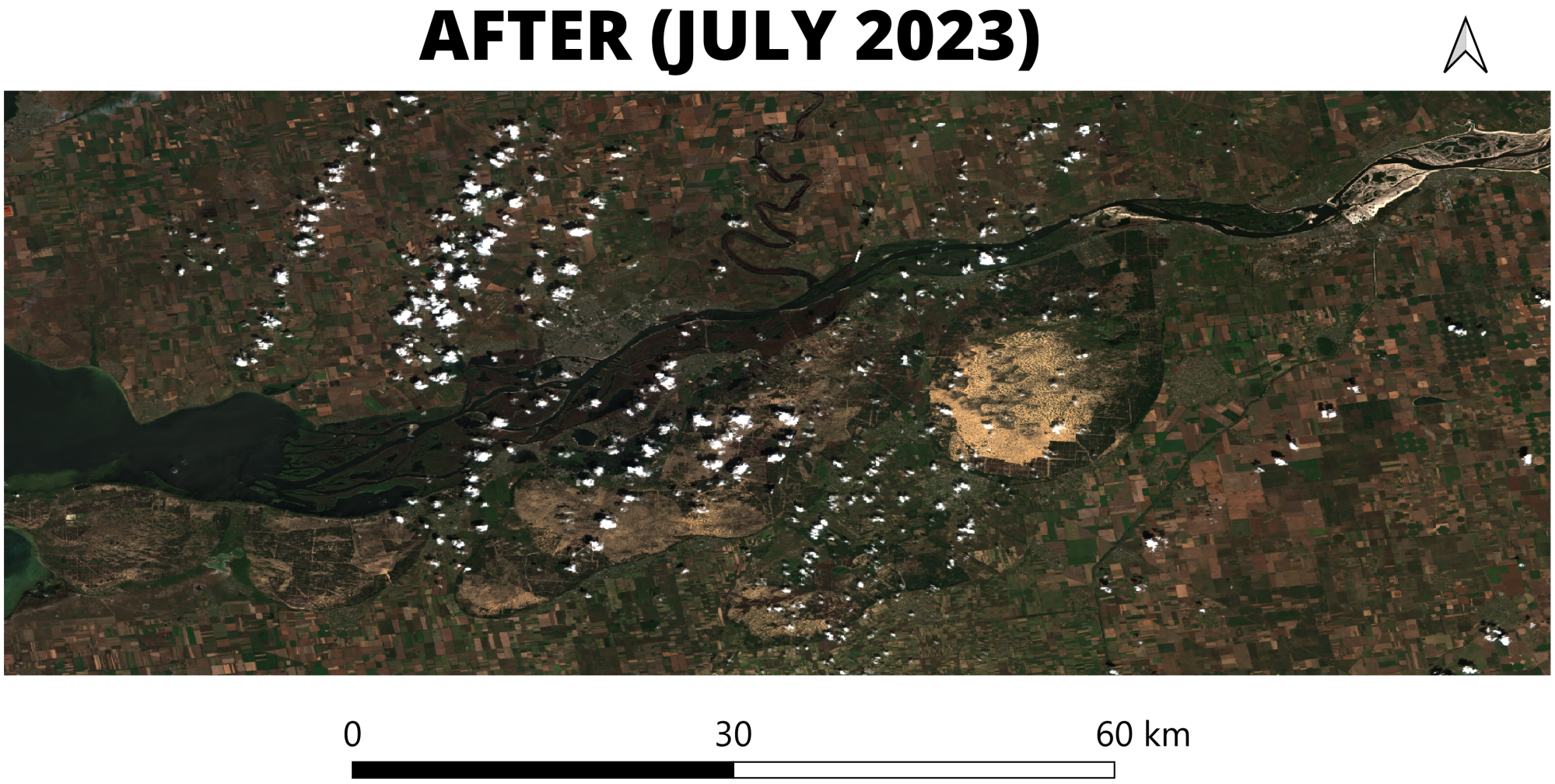
Treatment zone (Kherson Oblast, Ukraine) and the four Romanian control counties (Tulcea, Galati, Braila, Constanta) in their true geographic positions. Boundaries: GADM v4.1.

Figure S1-2 — Lower Dnipro before the breach (April-May 2023)



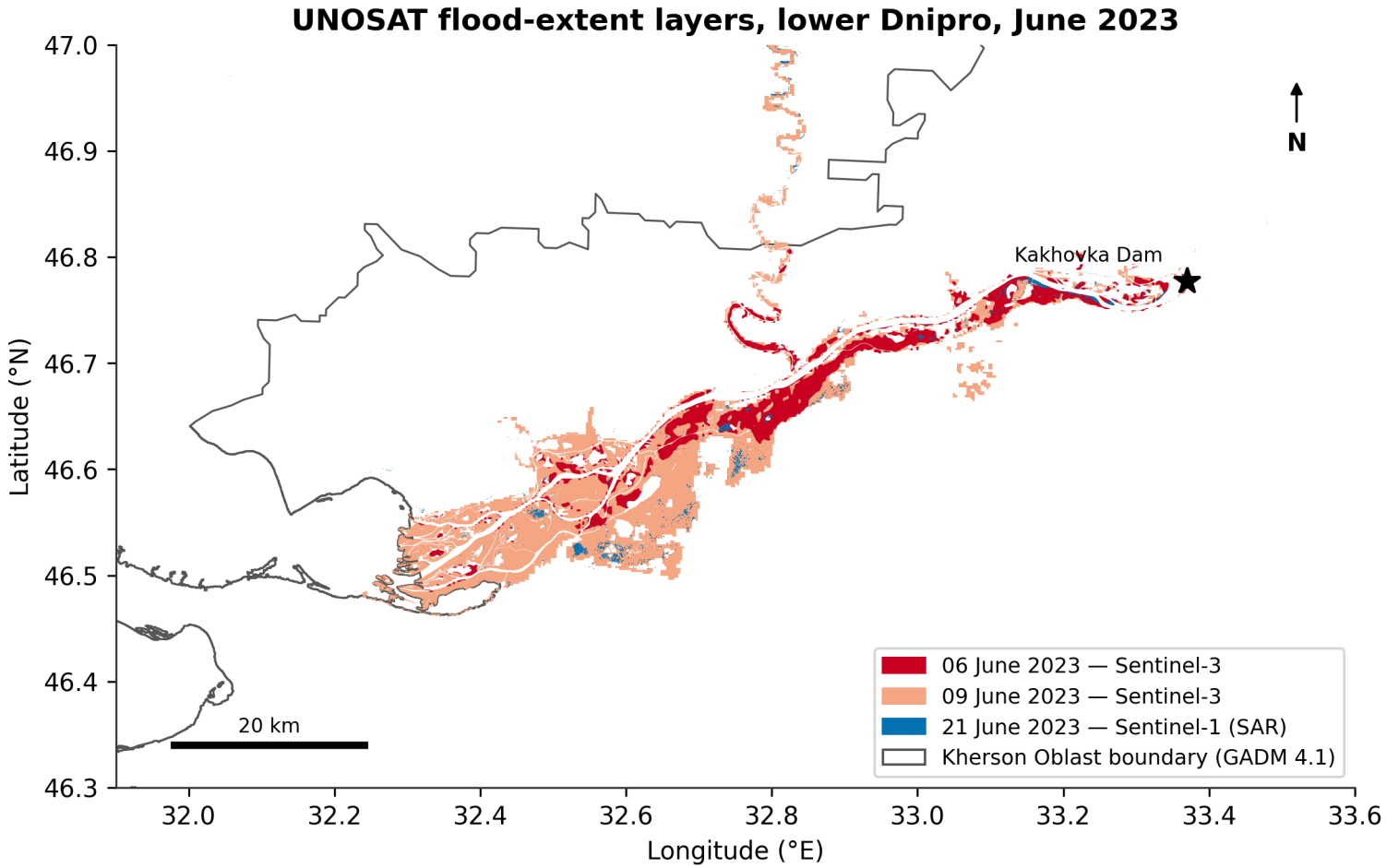
Sentinel-2 L2A true-colour mosaic, 1 April - 31 May 2023, least-cloud mosaicking, bbox 32.0-33.6E, 46.3-46.9N. The frame covers the downstream floodplain and only the south-western tip of the reservoir.

Figure S1-3 — Lower Dnipro after the breach (July 2023)



Same bbox and processing, July 2023. The drained reservoir tip near Nova Kakhovka is visible at top right. Context only; not used as statistical evidence.

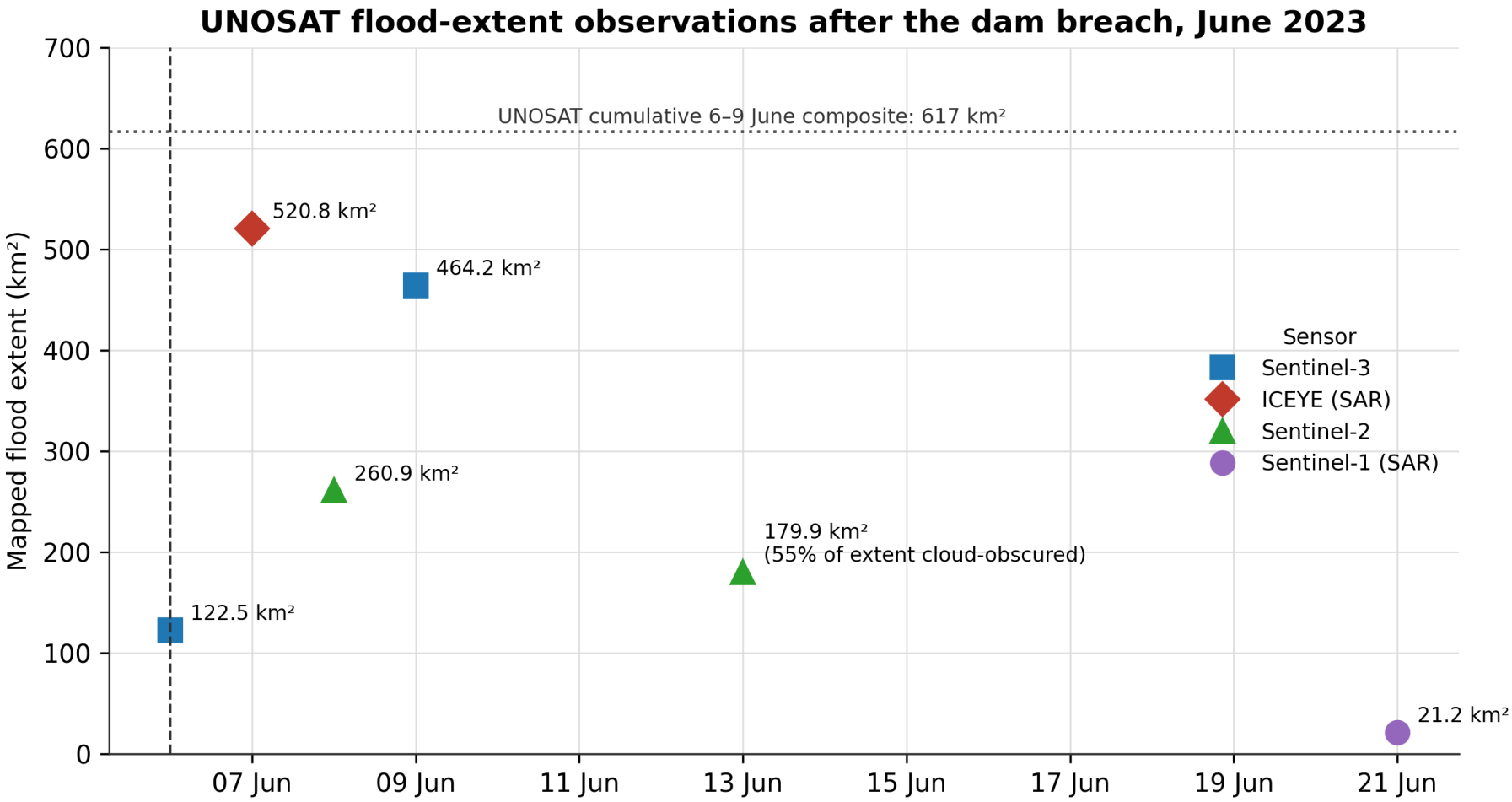
Figure S1-4 — UNOSAT flood-extent layers



Source: UNOSAT FL20230606UKR (preliminary, not field-validated). Layers come from different sensors and analysis extents.

UNOSAT FL20230606UKR layers for 6 June (Sentinel-3), 9 June (Sentinel-3) and 21 June (Sentinel-1). Preliminary, not field-validated; layers differ in sensor and analysis extent.

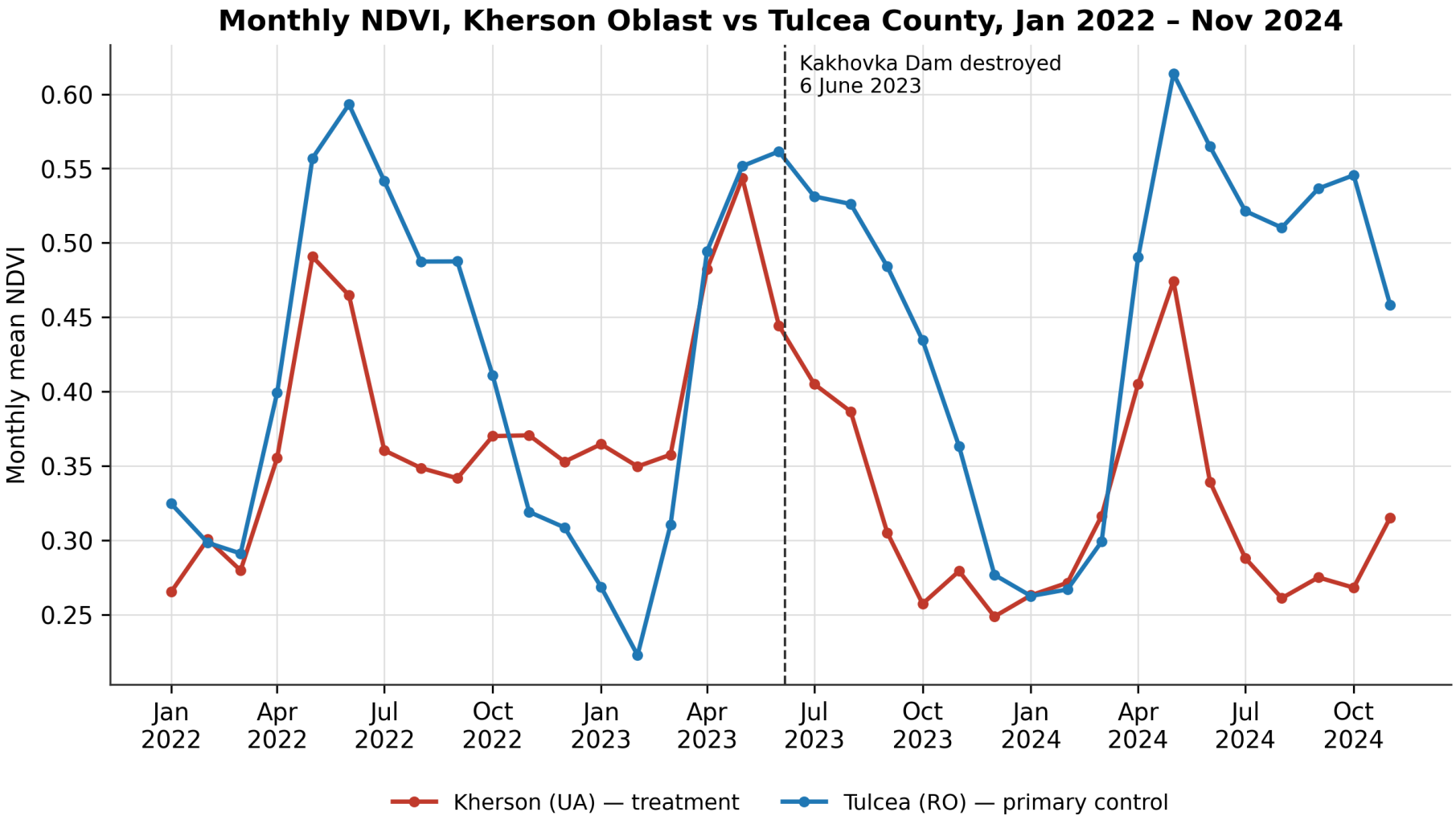
Figure S1-5 — UNOSAT flood-extent observations by sensor



Each point is a separate UNOSAT layer with its own sensor, resolution and analysis extent; values are not directly comparable across sensors.
Source: UNOSAT FL20230606UKR (preliminary, not field-validated).

Mapped flood area in each UNOSAT layer, labelled by sensor. Separate observations, not a continuous series; see outputs/flood_extent_table.csv.

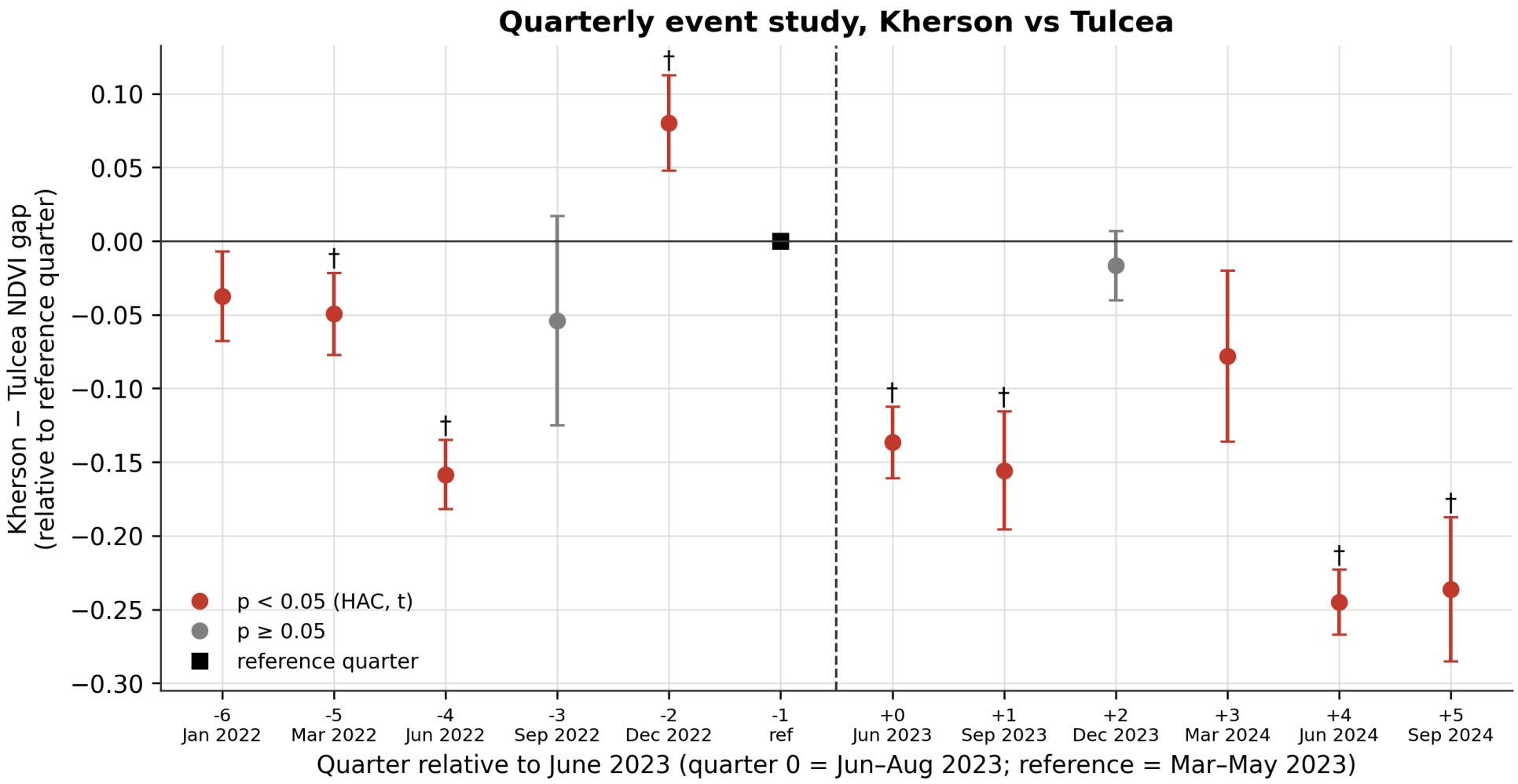
Figure S1-6 — Monthly NDVI, Kherson vs Tulcea



Sentinel-2 L2A via Sentinel Hub Statistical API; GADM polygons; cloud, shadow, cirrus, snow and water masked; per-pixel monthly median on a 0.002° grid. Descriptive only.

Monthly mean NDVI (Sentinel-2, GADM polygons, SCL masking), January 2022 - November 2024. Descriptive.

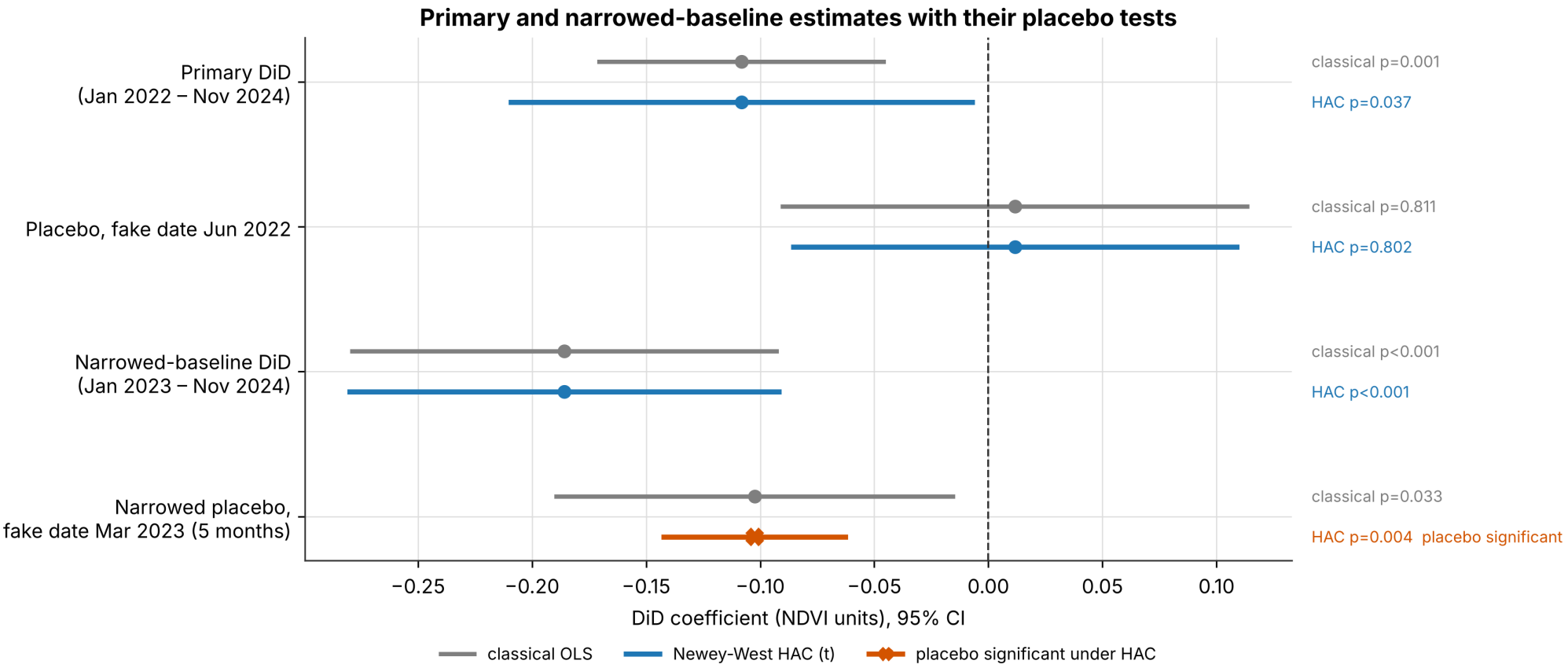
Figure S1-7 — Quarterly event study



† survives Bonferroni correction across 11 quarters ($p < 0.00455$). 95% CIs, Newey-West HAC (maxlags=3).

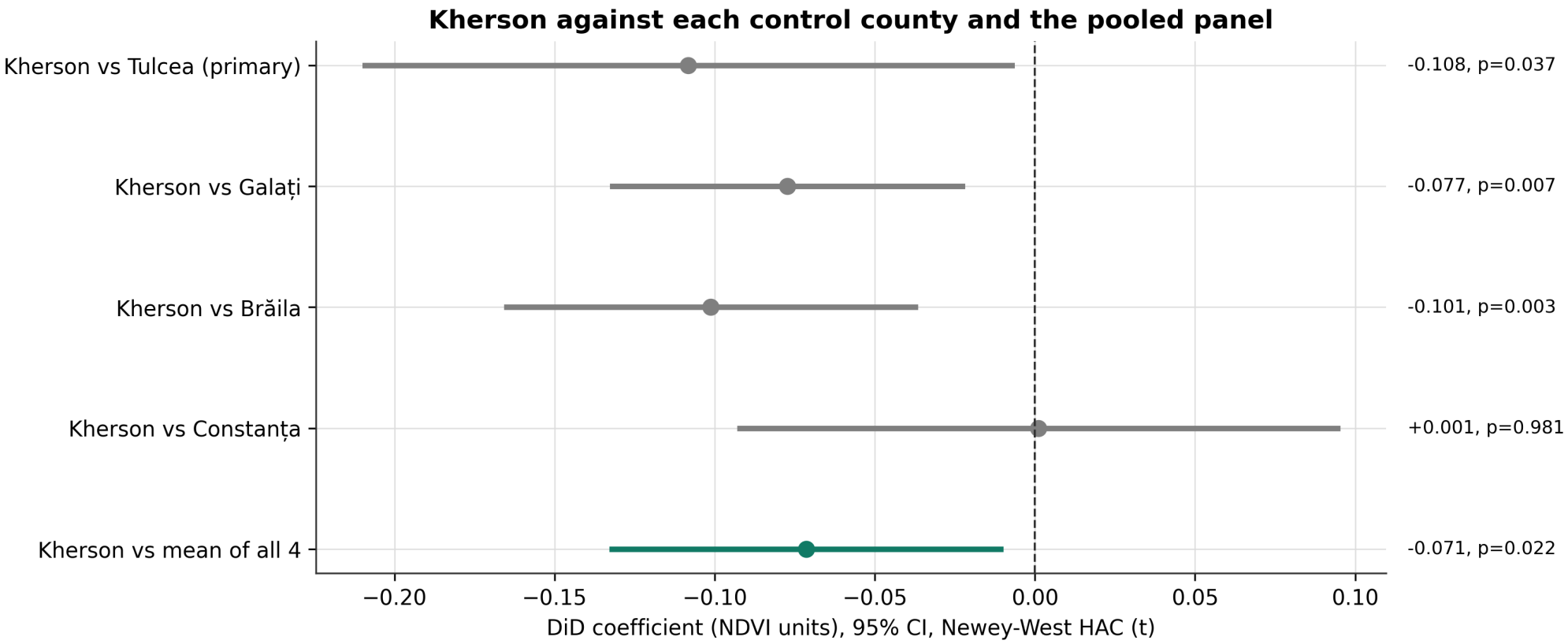
Kherson minus Tulcea NDVI gap by quarter relative to June 2023 (reference Mar-May 2023), Newey-West HAC 95% CIs; dagger marks quarters surviving Bonferroni correction.

Figure S1-8 — Primary and narrowed-baseline estimates with placebos



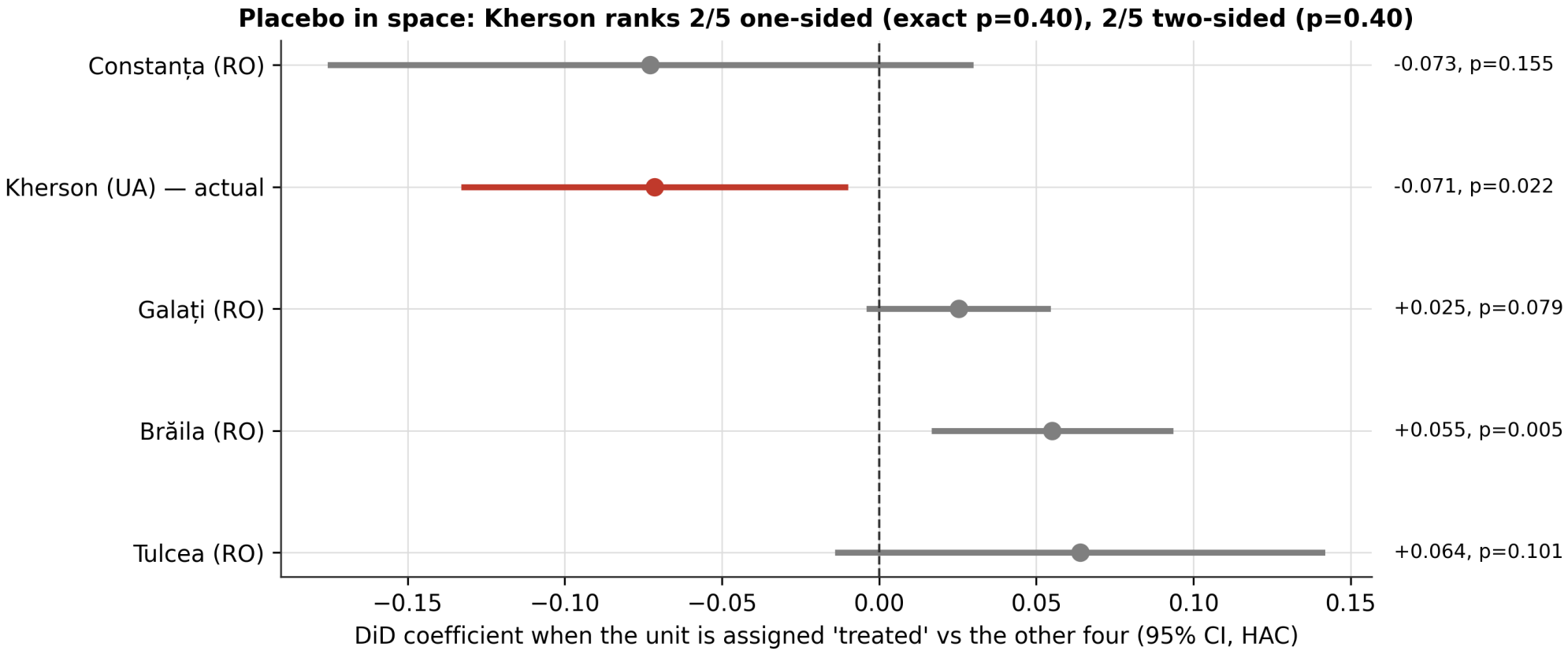
Classical and Newey-West HAC 95% CIs. Values from outputs/model_results.json.

Figure S1-9 — Kherson against each control and pooled



DiD estimates against each Romanian control county and their mean, Newey-West HAC 95% CIs.

Figure S1-10 — Placebo in space



Each of the five units assigned 'treated' in turn against the other four; exact randomization p-values from Kherson's rank.